

# Control and Modulation of the Stacked Polyphase Bridges Inverter

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**Abstract**—In order to meet the requirements of future electric vehicle (EV) drive systems, new converter topologies are required. The stacked polyphase bridges (SPB) inverter is a new topology which is well adapted to the needs for future EVs [1]. In addition to low power losses, the SPB inverter also has fault tolerance ability and good dynamic behavior. This paper presents a further analysis of the control and modulation concept of the SPB inverter. Simulation results are used to verify the analysis conclusions.

## I. INTRODUCTION

A conventional electric drive used in EV applications usually consists of a battery, a dc-link capacitor, a voltage source inverter (VSI), and a three-phase electric machine. There are three main requirements for the future drive system, which are reliability improvement, cost reduction and size (weight and volume) reduction. The SPB inverter is a new modular inverter topology based on the dc-side cascading of conventional two-level three-phase inverters. In [1], the SPB inverter was compared to another relatively recently proposed topology the modular high frequency converter [2] with regard to power dissipation during operation and the required silicon area. A control scheme was devised for the SPB inverter in order to get an indication of its ability to balance cell capacitor voltages and handle transient conditions. The calculation and simulation results imply that the SPB inverter has sufficiently high efficiency and can handle a fault in a submodule without interrupting operation. Thus, it is well suited for realising the future integrated electric drive system in terms of reliability improvement.

Some of the most bulky components in the conventional two-level VSI-based drive system are the passive components, especially the dc-link capacitor. Due to the switching operation, the inverter generates ripple currents in the dc-link current, thus a large dc-link capacitor is needed to smooth the battery current by absorbing the ripple currents in the dc link and to maintain a stable dc voltage within acceptable limits, otherwise the ripple will reduce the battery life. In a typical 60 kW HEV inverter, a dc-link capacitor of about 3.13 J/kW is required to prevent the large ripple current from flowing into the battery [3]. Therefore, minimizing the capacitor size by significantly reducing the ripple currents and increasing their frequency could help achieve a smaller and cheaper drive

system.

This paper shows that the SPB inverter can reduce the ripple current and has good dynamic behavior even with very small passive components, both by theoretical analysis and simulations. The reduction of the ripple is based on the phase shifted carrier modulation method. The paper is outlined as follows. In Section II, the SPB inverter and its main features are described. Section III gives a detailed system analysis in the frequency domain and proposes a modulation method for harmonic reduction in the SPB inverter. Braking case of the SPB-based EV is studied in Section IV. Simulation of the SPB inverter is carried out in Section V along with results validating the analysis in the previous section. Finally, conclusions are made in Section VI.

## II. BASIC FEATURES OF THE STACKED POLYPHASE BRIDGES INVERTER

The SPB inverter, shown in Fig. 1, consists of a series connected submodules and each submodule consists of a three-phase bridge with its dc-link capacitor. The ac terminal of each submodule is connected to one set of three-phase windings of a multi-star permanent magnet synchronous machine (PMSM) [4]. The integrated drive system can eliminate the connection cable between the SPB inverter and motor.

The dynamics of the battery current can be expressed as

$$L \frac{di_b}{dt} = E - Ri_b - \sum_{k=1}^N v_{\text{conv},k} \quad (1a)$$

$$C_{\text{conv},k} \frac{dv_{\text{conv},k}}{dt} = i_b - i_{\text{conv},k} \quad (1b)$$

where  $E$  is the battery voltage,  $i_b$  is the battery current,  $L$  and  $R$  are the total battery and cable inductance and resistance, respectively.  $v_{\text{conv},k}$ ,  $i_{\text{conv},k}$ , and  $C_{\text{conv},k}$  are input voltage, current and capacitance of submodule  $k$ , respectively.  $N$  is the total number of the submodules.

The SPB inverter could offer several significant advantages in compact EV and HEV applications. For instance, its cascaded nature allows for the use of low voltage semiconductor devices which permits high switching frequency. Another benefit is, due to its modularity, the SPB inverter can implement cell-level redundancy.

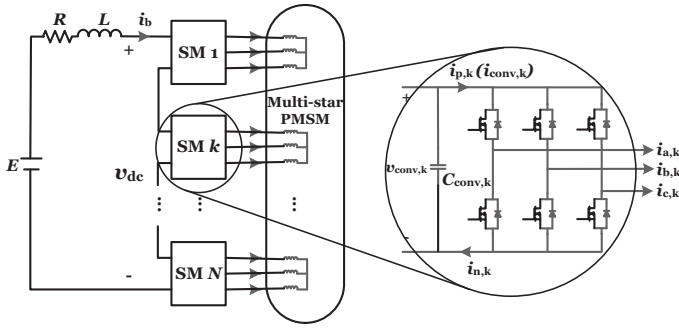


Fig. 1. Structure of the SPB inverter topology.

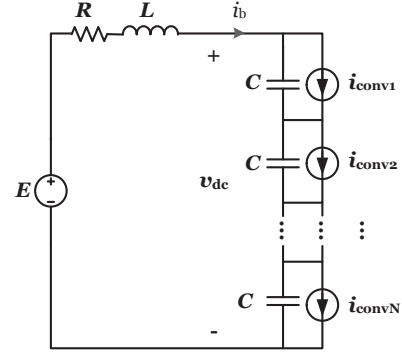


Fig. 2. An equivalent RLC circuit of the SPB inverter ( $N$  identical cells).

### III. HARMONICS CONTROL OF THE BATTERY CURRENT

Considering the cell structure in Fig. 1, let  $s_{x,k}$  be the time-varying binary switching function of phase leg  $x$  of cell  $k$  defined as:

$$s_{x,k} = \begin{cases} 1 & \text{the upper switch is on, the lower one is off,} \\ -1 & \text{the lower switch is on, the upper one is off.} \end{cases} \quad (2)$$

Then the positive and negative dc rail current of inverter  $k$  can be written as:

$$i_{p,k} = \frac{1}{2}[(s_{a,k} + 1)i_{a,k} + (s_{b,k} + 1)i_{b,k} + (s_{c,k} + 1)i_{c,k}] \quad (3a)$$

$$i_{n,k} = \frac{1}{2}[(s_{a,k} - 1)i_{a,k} + (s_{b,k} - 1)i_{b,k} + (s_{c,k} - 1)i_{c,k}] \quad (3b)$$

where

$$i_{a,k} = I_0 \cos(\omega_{0,k}t + \theta_{0,k}) \quad (4a)$$

$$i_{b,k} = I_0 \cos(\omega_{0,k}t + \frac{2\pi}{3} + \theta_{0,k}) \quad (4b)$$

$$i_{c,k} = I_0 \cos(\omega_{0,k}t - \frac{2\pi}{3} + \theta_{0,k}). \quad (4c)$$

Since we assume three-phase symmetric ac currents ( $i_{a,k} + i_{b,k} + i_{c,k} = 0$ ) in the analysis, equations (3) may be rewritten as:

$$i_{p,k} = i_{n,k} = \frac{1}{2}(s_{a,k}i_{a,k} + s_{b,k}i_{b,k} + s_{c,k}i_{c,k}). \quad (5)$$

Therefore, only common mode dc current needs to be considered.

Rename  $i_{p,k}$  as  $i_{\text{conv},k}$  and consider the system in Fig. 1. For the sake of simplicity, all capacitances are assumed to be equal to  $C$ , and current sources  $i_{\text{conv},k}$  are used to replace the three-phase bridges and the load. Based on these assumptions, the equivalent RLC circuit in Fig. 2 is obtained. By using superposition to analyze this circuit, we obtain the expression for  $i_b$  at frequency  $\omega$ :

$$I_b(\omega) = \frac{\sum_{k=1}^N I_{\text{conv},k}(\omega)}{N - \omega^2 LC + j\omega CR} \quad (6)$$

where  $I_b(\omega)$  and  $I_{\text{conv},k}(\omega)$  represent the amplitude of  $i_b$  and  $i_{\text{conv},k}$  at frequency  $\omega$ , respectively.

It is obvious that  $i_b$  is related to the summation  $\sum_{k=1}^N i_{\text{conv},k}$ . If some of the harmonic components in the sum can be cancelled out, the same order harmonics in  $i_b$  will also disappear. The analysis in this paper will be performed in the frequency domain using double Fourier series expansion. The general form of the switching function  $s_{a,k}$  is periodic in both carrier wave angular frequency  $\omega_{c,k}$  and the fundamental frequency  $\omega_{0,k}$ . Therefore, it can be expanded into the double Fourier series [5]:

$$\begin{aligned} s_{a,k}(t) = & \frac{A_{00}}{2} \\ & + \sum_{n=1}^{\infty} [A_{0n} \cos(n[\omega_{0,k}t + \theta_{0,k}]) + B_{0n} \sin(n[\omega_{0,k}t + \theta_{0,k}])] \\ & + \sum_{m=1}^{\infty} [A_{m0} \cos(m[\omega_{c,k}t + \theta_{c,k}]) + B_{m0} \sin(m[\omega_{c,k}t + \theta_{c,k}])] \\ & + \sum_{m=1}^{\infty} \sum_{n=-\infty}^{+\infty} [A_{mn} \cos(m[\omega_{c,k}t + \theta_{c,k}] + n[\omega_{0,k}t + \theta_{0,k}]) \\ & + B_{mn} \sin(m[\omega_{c,k}t + \theta_{c,k}] + n[\omega_{0,k}t + \theta_{0,k}])] \end{aligned} \quad (7)$$

where  $m$  and  $n$  are the carrier and sideband index variables, and  $\theta_{c,k}$  and  $\theta_{0,k}$  are initial angles of carrier and reference waveforms of submodule  $k$ ,  $A_{mn}$  and  $B_{mn}$  are the real and imaginary part of the complex Fourier coefficient.

Multiplication in the time domain is equivalent to convolution in the frequency domain. Hence, it is clear that if the time-varying signals in (5) are known, the spectra of their time-varying product can be obtained by using convolution in the frequency domain

$$I_{\text{conv},a,k} = \frac{1}{2}(S_{a,k} \otimes I_{a,k}) \quad (8a)$$

$$I_{\text{conv},b,k} = \frac{1}{2}(S_{b,k} \otimes I_{b,k}) \quad (8b)$$

$$I_{\text{conv},c,k} = \frac{1}{2}(S_{c,k} \otimes I_{c,k}) \quad (8c)$$

$$I_{\text{conv},k} = I_{\text{conv},a,k} + I_{\text{conv},b,k} + I_{\text{conv},c,k} \quad (8d)$$

where  $I_{a,k}$ ,  $I_{b,k}$  and  $I_{c,k}$  are spectra of the corresponding phase currents,  $S_{a,k}$ ,  $S_{b,k}$  and  $S_{c,k}$  are spectra of the corresponding

switching functions, and  $I_{\text{conva},k}$ ,  $I_{\text{convb},k}$  and  $I_{\text{convc},k}$  are spectra of the corresponding currents going through the phase legs. The symbol  $\otimes$  denotes two-dimensional convolution of the Fourier coefficients of the signals, i.e.

$$(X \otimes Y)^{(m,n)} = \sum_{k_m=-\infty}^{+\infty} \sum_{k_n=-\infty}^{+\infty} X^{(k_m,k_n)} Y^{(m-k_m,n-k_n)}. \quad (9)$$

Since the spectrum of the phase current only contains fundamental component, which means that only the components where  $m = 0, n = \pm 1$  need to be considered in the convolution, the following equation can be obtained from (8) and (9):

$$\begin{aligned} I_{\text{conva},k}^{(m,n)} &= \frac{1}{2} \left[ \sum_{k_m=-\infty}^{+\infty} \sum_{k_n=-\infty}^{+\infty} I_{a,k}^{(k_m,k_n)} S_{a,k}^{(m-k_m,n-k_n)} \right] \\ &= \frac{1}{2} [I_{a,k}^{(0,-1)} S_{a,k}^{(m,n+1)} + I_{a,k}^{(0,1)} S_{a,k}^{(m,n-1)}], \end{aligned} \quad (10)$$

where  $I_{a,k}^{(0,-1)} = \frac{I_0}{2} e^{j\theta_{0,k}}$ ,  $I_{a,k}^{(0,1)} = \frac{I_0}{2} e^{-j\theta_{0,k}}$ .

The above equation gives the spectral content of the contribution of one phase-leg to the dc-link current. It can be seen that  $I_{\text{conva},k}$  is simply the superposition of the phase-leg's switching function spectrum with phase shifts of  $\pm \omega_{0,k}$  [6]. Ordinary naturally sampled sine-triangle modulation will be used for the further analysis. The harmonic coefficients of the switching function generated by this modulation method are [7]:

$$S_{a,k}^{(0,0)} = 0 \quad (11a)$$

$$S_{a,k}^{(0,\pm 1)} = M/2 \quad (11b)$$

$$S_{a,k}^{(m,n)} = \frac{2}{m\pi} J_n(mM \frac{\pi}{2}) \sin((m+n) \frac{\pi}{2}), \quad m > 1 \quad (11c)$$

where  $M$  is the modulation index,  $J_n$  is the Bessel function of order  $n$ .

Substituting the equation set (11) into (10) and identifying the coefficients of the double Fourier series expansion for  $i_{\text{conva},k}$  the following is obtained:

$$A_{00} = \frac{MI_0}{4} \cos \theta_{0,k} \quad (12a)$$

$$A_{01} = \frac{I_0}{2} \cos \theta_{0,k}, \quad A_{02} = \frac{MI_0}{4} \cos \theta_{0,k} \quad (12b)$$

$$B_{01} = \frac{-I_0}{2} \sin \theta_{0,k}, \quad B_{02} = \frac{-MI_0}{4} \sin \theta_{0,k} \quad (12c)$$

$$A_{0n} = B_{0n} = 0, \quad n \geq 2 \quad (12d)$$

$$\begin{aligned} A_{mn} &= \frac{I_0}{m\pi} \cos \theta_{0,k} \cos((m+n) \frac{\pi}{2}) \\ &\times [J_{n+1}(m \frac{\pi}{2} M) - J_{n-1}(m \frac{\pi}{2} M)] \end{aligned} \quad (12e)$$

$$\begin{aligned} B_{mn} &= \frac{I_0}{m\pi} \sin \theta_{0,k} \cos([m+n] \frac{\pi}{2}) \\ &\times [-J_{n+1}(m \frac{\pi}{2} M) - J_{n-1}(m \frac{\pi}{2} M)]. \end{aligned} \quad (12f)$$

The switched phase-leg currents  $i_{\text{convb},k}$  and  $i_{\text{convc},k}$  are obtained by the same way, then by substituting them into (8d)

the coefficients of the dc-link current  $i_{\text{conv},k}$  of cell  $k$  can be obtained:

$$A_{00} = \frac{3MI_0}{4} \cos \theta_{0,k} \quad (13a)$$

$$A_{0n} = B_{0n} = 0, \quad n \geq 1 \quad (13b)$$

$$\begin{aligned} A_{mn} &= \frac{I_0}{m\pi} \cos \theta_{0,k} \cos((m+n) \frac{\pi}{2}) [1 + 2 \cos(\frac{2n\pi}{3})] \\ &\times [J_{n+1}(m \frac{\pi}{2} M) - J_{n-1}(m \frac{\pi}{2} M)] \end{aligned} \quad (13c)$$

$$\begin{aligned} B_{mn} &= \frac{I_0}{m\pi} \sin \theta_{0,k} \cos((m+n) \frac{\pi}{2}) [1 + 2 \cos(\frac{2n\pi}{3})] \\ &\times [-J_{n+1}(m \frac{\pi}{2} M) - J_{n-1}(m \frac{\pi}{2} M)]. \end{aligned} \quad (13d)$$

From (13a) and (13b), it is obvious that the summation of the contributions from all the three-phase legs results in the cancellation of the fundamental and the second order harmonic but preserves the dc component. Considering  $\cos((m+n) \frac{\pi}{2})$  in (13c),  $A_{mn}$  is nonzero only when  $m+n$  is even; considering  $1 + 2 \cos(\frac{2n\pi}{3})$ ,  $A_{mn}$  is nonzero only when  $n = \pm 3p, p = 1, 2$ , etc. Thus, if  $m$  is odd,  $A_{mn}$  is nonzero only when  $n = \pm 6p - 3$ ; if  $m$  is even,  $A_{mn}$  is nonzero only when  $n = \pm 6p$ . The same conclusions hold for  $B_{mn}$  in (13d). The carrier harmonics and the corresponding sidebands of  $i_{\text{conv},k}$  concluded from (13) are listed in Table I,  $p = 1, 2$ , etc.

Summing the dc-link currents  $i_{\text{conv},k}$  of all submodules, the Fourier coefficients of the summation  $\sum_{k=1}^N i_{\text{conv},k}$  can be obtained as:

$$\sum_{k=1}^N A_{00} = \frac{3MNI_0}{4} \cos \theta_{0,k} \quad (14a)$$

$$\sum_{k=1}^N A_{0n} = B_{0n} = 0, \quad n \geq 1 \quad (14b)$$

$$\begin{aligned} \sum_{k=1}^N A_{mn} &= \frac{I_0}{m\pi} \cos((m+n) \frac{\pi}{2}) [1 + 2 \cos(\frac{2n\pi}{3})] \\ &\times \{ \cos \theta_{0,k} [J_{n+1}(m \frac{\pi}{2} M) - J_{n-1}(m \frac{\pi}{2} M)] \sum_{k=1}^N \cos m\theta_{c,k} \\ &+ \sin \theta_{0,k} [-J_{n+1}(m \frac{\pi}{2} M) - J_{n-1}(m \frac{\pi}{2} M)] \sum_{k=1}^N \sin m\theta_{c,k} \} \end{aligned} \quad (14c)$$

$$\begin{aligned} \sum_{k=1}^N B_{mn} &= \frac{I_0}{m\pi} \cos((m+n) \frac{\pi}{2}) [1 + 2 \cos(\frac{2n\pi}{3})] \\ &\times \{ \cos \theta_{0,k} [J_{n+1}(m \frac{\pi}{2} M) - J_{n-1}(m \frac{\pi}{2} M)] \sum_{k=1}^N (-\sin m\theta_{c,k}) \\ &+ \sin \theta_{0,k} [-J_{n+1}(m \frac{\pi}{2} M) - J_{n-1}(m \frac{\pi}{2} M)] \sum_{k=1}^N (\cos m\theta_{c,k}) \}. \end{aligned} \quad (14d)$$

From (14a) and (14b), it can be seen that there is only dc component in the summation  $\sum_{k=1}^N i_{\text{conv},k}$ ; no fundamental

TABLE I  
CARRIER HARMONICS AND SIDEBANDS OF  $i_{\text{conv},k}$

| Carrier harmonics | Sidebands        |                  |
|-------------------|------------------|------------------|
| even              | when $m$ is odd  | $n = \pm 6p - 3$ |
|                   | when $m$ is even | $n = \pm 6p$     |

TABLE II  
CARRIER HARMONICS AND SIDEBANDS OF  $i_b$  ( $\theta_{c,k} = \frac{2\pi k}{N}$ )

| Carrier harmonics | when $N$ is odd<br>when $N$ is even           | $m = 2pN$<br>$m = pN$            |
|-------------------|-----------------------------------------------|----------------------------------|
| Sidebands         | when $m = pN$ is odd<br>when $m = pN$ is even | $n = \pm 6p - 3$<br>$n = \pm 6p$ |

or second order harmonic. Equations (14c) and (14d) show the real ( $\sum_{k=1}^N A_{mn}$ ) and imaginary ( $\sum_{k=1}^N B_{mn}$ ) Fourier coefficients of carrier harmonics and sideband harmonics of  $\sum_{k=1}^N i_{\text{conv},k}$ . According to the properties of trigonometric summations  $\sum_{k=1}^N (\sin \theta_{c,k})$  and  $\sum_{k=1}^N (\cos \theta_{c,k})$  are equal to zero when  $\theta_{c,k} = \frac{2\pi}{N} \times k$ . In order to have the most harmonics cancelled in  $\sum_{k=1}^N i_{\text{conv},k}$ ,  $\theta_{c,k} = \frac{2\pi}{N} \times k$  is chosen. Thus, carrier harmonics and the corresponding sidebands are cancelled when  $m \neq pN$ ,  $N$  is even; or  $m \neq 2pN$ ,  $N$  is odd. From the relationship of battery current  $i_b$  and the summation  $\sum_{k=1}^N i_{\text{conv},k}$  shown in (6), the spectrum of  $i_b$  has the same components as the spectrum of  $\sum_{k=1}^N i_{\text{conv},k}$ , which is shown in Table II.

#### IV. DYNAMIC BEHAVIOR

For future drive EV systems, less bulky capacitors are needed for a compact design. The ability to operate with small submodule capacitors is a big advantage of the SPB inverter. For the dimensioning of the submodule capacitor, the acceptable ripple in the dc-link voltage and overshoot in  $v_{dc}$  under transient conditions must be considered. As a result of the high switching frequency made possible by low voltage semiconductors, the size of the cell capacitors in the SPB inverter can be reduced. However, the value of the capacitance influences the transient behavior of the system under special conditions. In [1], it was shown that the SPB inverter displays good dynamic and static balancing of the capacitor voltages when one submodule fails and is bypassed. In this section, considering the braking case, the dynamic behavior of the SPB inverter with small cell capacitors is discussed.

Considering a multi-star PMSM as load and assuming there is a sudden drop in electrical torque reference at  $t = t_1$ , which is representative of the braking case, the energy can be dumped into the machine and dissipated in the stator windings by driving the torque reference from the q-axis to the d-axis. In Section III, the equivalent RLC circuit of an  $N$ -cell SPB inverter has been obtained. If the harmonics in the circuit are neglected, this  $N$ -cell RLC circuit could be seen as equivalent to one big cell, which is shown in Fig. 3. The transfer function

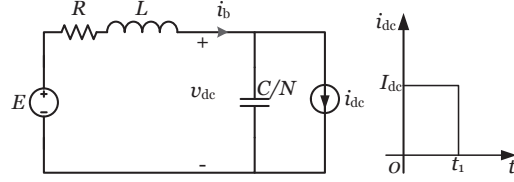


Fig. 3. An equivalent circuit of the SPB inverter when the harmonics in the circuit are neglected.

between  $\Delta v_{dc}$  and  $i_{dc}$  is obtained as:

$$\frac{\Delta V_{dc}(s)}{I_{dc}(s)} = \frac{-N(Ls + R)}{LCs^2 + RCs + N} \quad (15)$$

where  $\Delta V_{dc}(s)$  and  $I_{dc}(s)$  are Laplace-transformed  $\Delta v_{dc}$  and  $i_{dc}$ .

From (15), the time-varying expression for  $\Delta v_{dc}$  can be obtained as:

$$\Delta v_{dc}(t) = \begin{cases} W_0 + W_1 e^{p_1 t} + W_2 e^{p_2 t} & \text{if } \delta \geq 0 \\ V_0 + V_1 e^{(\alpha + j\beta)t} + V_2 e^{(\alpha - j\beta)t} & \text{if } \delta < 0 \end{cases} \quad (16)$$

where

$$\delta = \frac{R^2}{L^2} - \frac{4N}{LC} \quad (17a)$$

$$p_{1,2} = -\frac{R}{2L} \pm \frac{\sqrt{\delta}}{2}, \quad \alpha \pm j\beta = -\frac{R}{2L} \pm j\frac{\sqrt{|\delta|}}{2} \quad (17b)$$

$$W_0 = V_0 = -RI_{dc} \quad (17c)$$

$$W_1 = \frac{NI_{dc}(-R - L\sqrt{\delta})}{LC\delta - RC\sqrt{\delta}}, \quad W_2 = \frac{NI_{dc}(-R + L\sqrt{\delta})}{LC\delta + RC\sqrt{\delta}} \quad (17d)$$

$$V_1 = \frac{NI_{dc}(-R - jL\sqrt{|\delta|})}{-LC|\delta| - jRC\sqrt{|\delta|}}, \quad V_2 = \frac{NI_{dc}(-R + jL\sqrt{|\delta|})}{-LC|\delta| + jRC\sqrt{|\delta|}} \quad (17e)$$

By neglecting  $R$ , which could also give the correct value of the first swing of the capacitor voltage, an upper bound for how high the overvoltage may become during the braking case can be obtained as:

$$\Delta v_{dc}(t) = \hat{V}_1 e^{j\hat{\beta}t} + \hat{V}_2 e^{-j\hat{\beta}t} \quad (18)$$

where

$$\hat{\beta} = \sqrt{\frac{N}{LC}}, \quad \hat{V}_1 = j\frac{I_{dc}}{2}\sqrt{\frac{NL}{C}}, \quad \hat{V}_2 = -j\frac{I_{dc}}{2}\sqrt{\frac{NL}{C}} \quad (19)$$

According to (18) and (19), it is obvious that the overshoot in  $v_{dc}$  is related to the circuit parameters, especially the value of the passive components. Fig. 4 shows that the overshoot in  $v_{dc}$  increases with decreasing  $C$  for the given battery and cable model which will be introduced in Section V. From the calculation,  $\Delta v_{dc}$  in this case is less than 5% when the cell capacitance is bigger than  $0.065 \text{ J/kW}$ . Therefore, the SPB inverter can keep  $\Delta v_{dc}$  within acceptable limit even with fairly small cell capacitors.

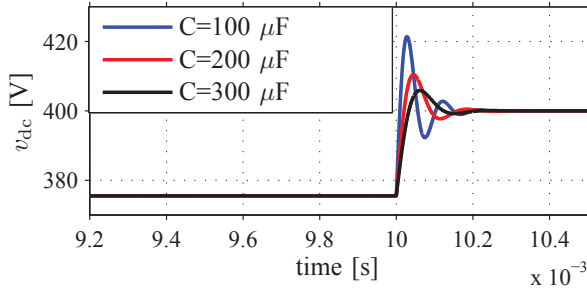


Fig. 4. The sum of cell voltages under different cell capacitances.

## V. SIMULATION

To investigate the characteristics of the SPB inverter with small passive components and the suggested modulation method, a simulation study is carried out in this section. A impedance-based 400 V battery and cable model is made according to [8] [9]. The total equivalent resistance and inductance of the battery and cable model are quite small, 0.35  $\Omega$  and 8  $\mu\text{H}$ , respectively. A high-speed PMSM drive system for an electric vehicle is chosen for the simulation, and the main parameters of the simulated system are shown in Table III.

According to Section III, a cell capacitor of 0.065 J/kW is needed to limit the ripple in the battery current less than 2%. However, considering a single conventional two-level inverter which has the same specification as the example SPB system, the switching frequency used in VSI of EV is typically between 5kHz and 10kHz [10]. A dc-link capacitor of 4.4 J/kW, which is 4.27 times as the sum of cell capacitances (1.03 J/kW) in the SPB inverter, is required to keep the same ripple limitation as that in the SPB inverter.

### A. Harmonic Reduction of the Battery Current

Based on the analysis presented in Section III, a simulation study was carried out for a PMSM drive in both the carrier phase-shift case and the carrier in-phase case to verify the effect of the modulation method on the reduction of ripple current. Fig. 5 and Fig. 6 show simulated summation current  $\sum_{k=1}^N i_{\text{conv},k}$ , total capacitor voltage  $v_{dc}$  and the battery current  $i_b$ . The carrier phase-shift modulation results a 22% ripple reduction in the sum of cell voltages and a significant elimination of high frequency harmonics in the battery current.

TABLE III  
PARAMETERS USED FOR SIMULATION

| Name                        | Value | Unit | Symbol              |
|-----------------------------|-------|------|---------------------|
| Battery voltage             | 400   | V    | $E$                 |
| Number of cells             | 4     | -    | $N$                 |
| Cell capacitor voltage      | 94    | V    | $v_{\text{conv},k}$ |
| Switching frequency         | 100   | kHz  | $f_{\text{sw}}$     |
| Rated power                 | 30    | kW   | $P$                 |
| Motor nom. stator frequency | 1000  | Hz   | $f_1$               |

The spectra of the battery current in the two cases are shown in Fig. 7 and Fig. 8. In these figures, on the  $y$ -axes are the amplitudes of the harmonic components normalized based on the dc component and on the  $x$ -axes are harmonic orders. From Fig. 7, when  $\theta_{c,k} = 0$ , the carrier harmonic and its sidebands of  $i_b$  appear at  $m = p \times f_{\text{sw}}/f_1$  for  $p = 1, 2$ , etc. However, in Fig. 8, when  $\theta_{c,k} = k\pi/2$ , the carrier harmonic and its sidebands of  $i_b$  only appear at  $m = 4p$  for  $p = 1, 2$ , etc. The results are in accordance with the conclusion in Section III. Moreover, it can be seen that the distortion of the battery current is fairly small for both cases.

### B. Simulation of Sudden Reduction of Electrical Torque

A simulation model of the described vehicle braking case in Section IV was built. The results of simulations are compared

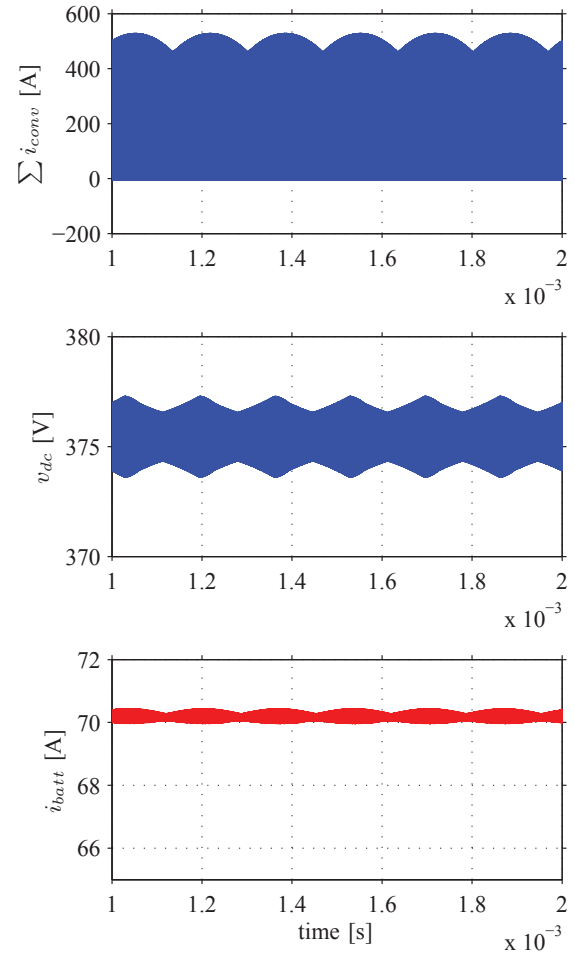


Fig. 5. Simulated waveforms for the SPB inverter. There is no phase shift in triangle carriers. Top: Sum of  $i_{\text{conv},k}$ . Middle: Sum of cell voltages. Bottom: Battery current.

TABLE IV  
DIMENSION OF SUBMODULE CAPACITANCE

|                            | C=250 $\mu\text{F}$ | C=110 $\mu\text{F}$ |
|----------------------------|---------------------|---------------------|
| Calculated $\Delta V_{dc}$ | 7.6 V (1.9%)        | 17.5 V (4.3%)       |
| Simulated $\Delta V_{dc}$  | 6.9 V (1.7%)        | 19.2 V (4.8%)       |

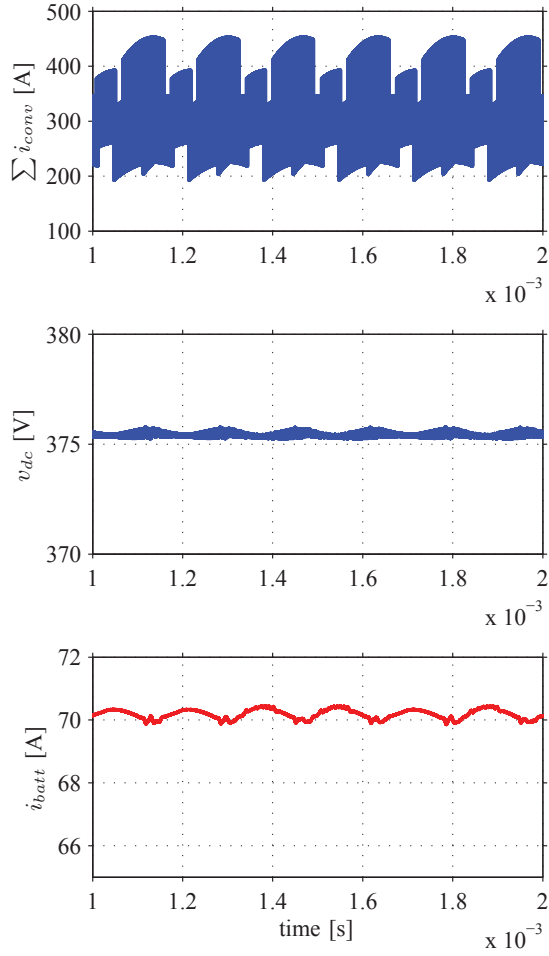


Fig. 6. Simulated waveforms for the SPB inverter. There is 90 electrical degrees phase shift in triangle carriers. Top: Sum of  $i_{conv,k}$ . Middle: Sum of cell voltages. Bottom: Battery current.

to the results from (15) in Table IV. As can be seen, the simulations show a good agreement with the theoretical calculations. In Fig. 9 and 10 simulated waveforms are displayed. At  $t=0$  ms the system is started and the cell capacitor voltages are well balanced. At  $t=2$  ms, there is a sudden reduction of electrical torque  $T_e$ , emulating the braking case. The overshoot in  $v_{dc}$  is approximately 5% and 2% by using cell capacitors of  $0.065 J/kW$  ( $110 \mu F$ ) and  $0.15 J/kW$  ( $250 \mu F$ ), respectively. The system enters steady state in 0.5 ms.

## VI. CONCLUSION

This paper presents an analysis of the SPB converter with small dc-side passive components. The effects of modulation schemes on the battery current were analyzed. The results show that the carrier harmonic and its sidebands of the battery current can be cancelled by properly phase shifting the initial angle of carriers among submodules. The reduction of the ripple currents makes the usage of smaller passive components in the dc-side feasible. However, there is very small ripple in the battery current even without phase shifting carriers,

which indicates that the SPB inverter can still have good performance under asynchronous communication situation. In addition, under the braking condition of the EV, the overshoot in the overall capacitor voltages is still in a safe range even with small dc-side passive components. Compared to the conventional two-level three-phase converter, the weight and volume of passive components can likely be significantly reduced, which is critical for integrated drive systems.

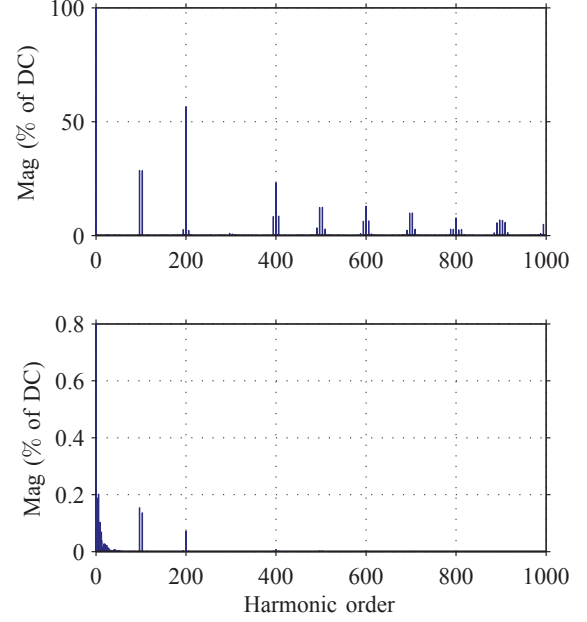


Fig. 7. Spectrum analysis. There is no phase shift in triangle carriers. Top: Sum of  $i_{conv,k}$ . Bottom: Battery current.

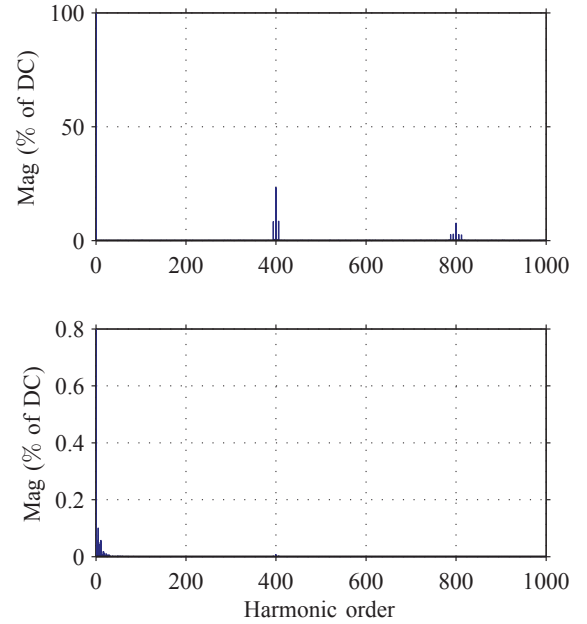


Fig. 8. Spectrum analysis. There is 90 electrical degrees phase shift in triangle carriers. Top: Sum of  $i_{conv,k}$ . Bottom: Battery current.

An important issue for the future work is the experimental verification of the control scheme and modulation method. A prototype is currently being built, Fig. 11 shows one cell. This work will also establish further requirements regarding capacitor size and communication among cells.

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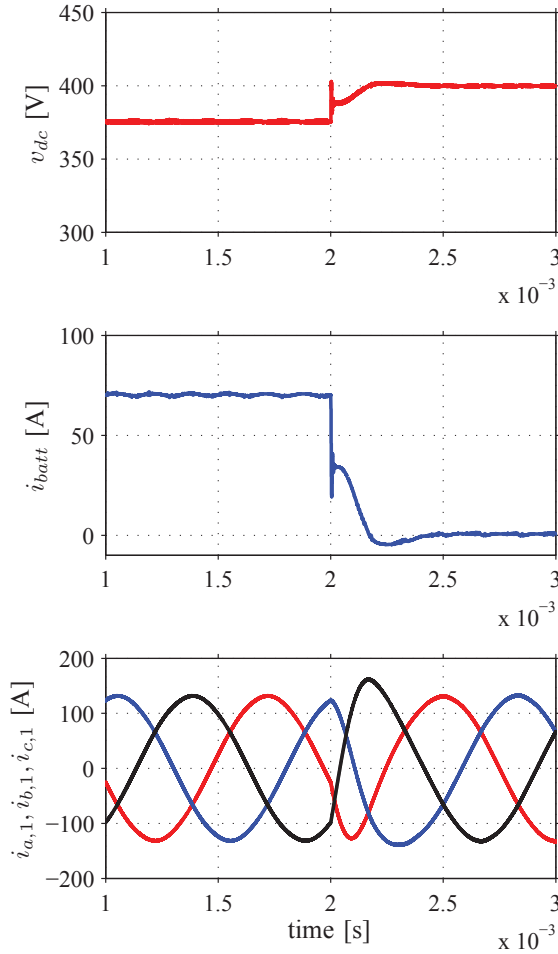


Fig. 9. Simulated waveforms for the SPB inverter. There is a sudden reduction in  $T_e$  ( $C=250 \mu\text{F}$ ,  $0.15 \text{ J/kW}$ ) at  $t=2 \text{ ms}$ . Top: Sum of cell voltages. Middle: Battery current. Bottom: Stator currents of one cell.

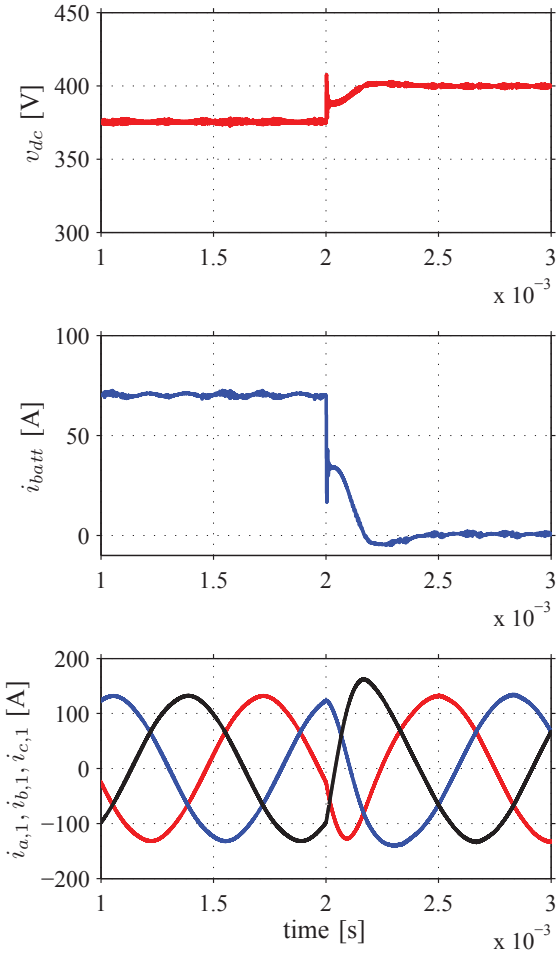


Fig. 10. Simulated waveforms for the SPB inverter. There is a sudden reduction in  $T_e$  ( $C=110 \mu\text{F}$ ,  $0.065 \text{ J/kW}$ ) at  $t=2 \text{ ms}$ . Top: Sum of cell voltages. Middle: Battery current. Bottom: Stator currents of one cell.

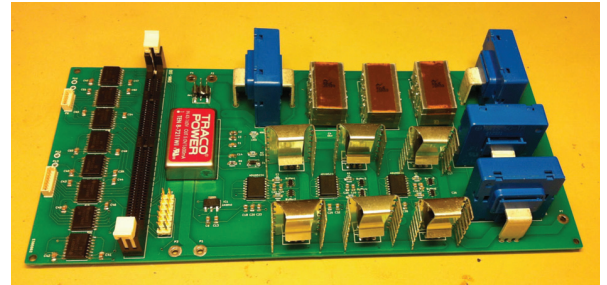


Fig. 11. PCB board of one cell.